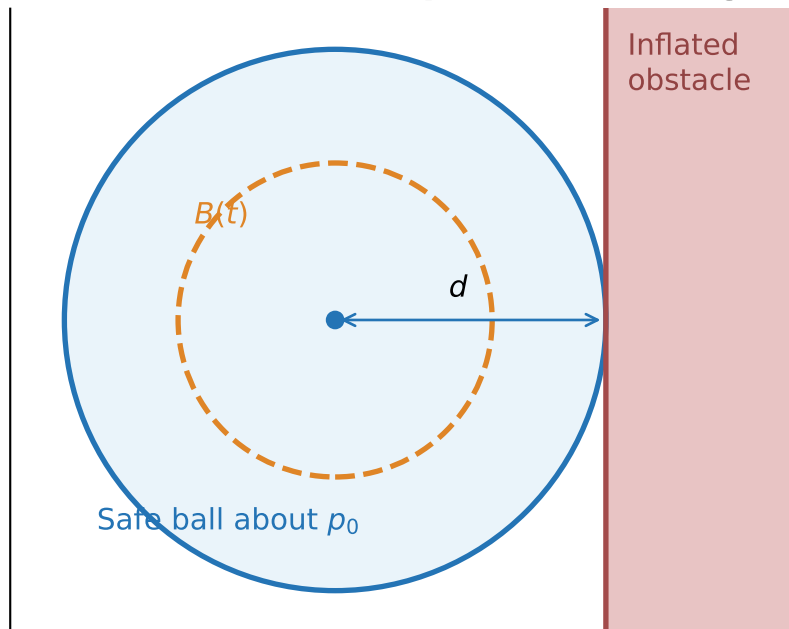


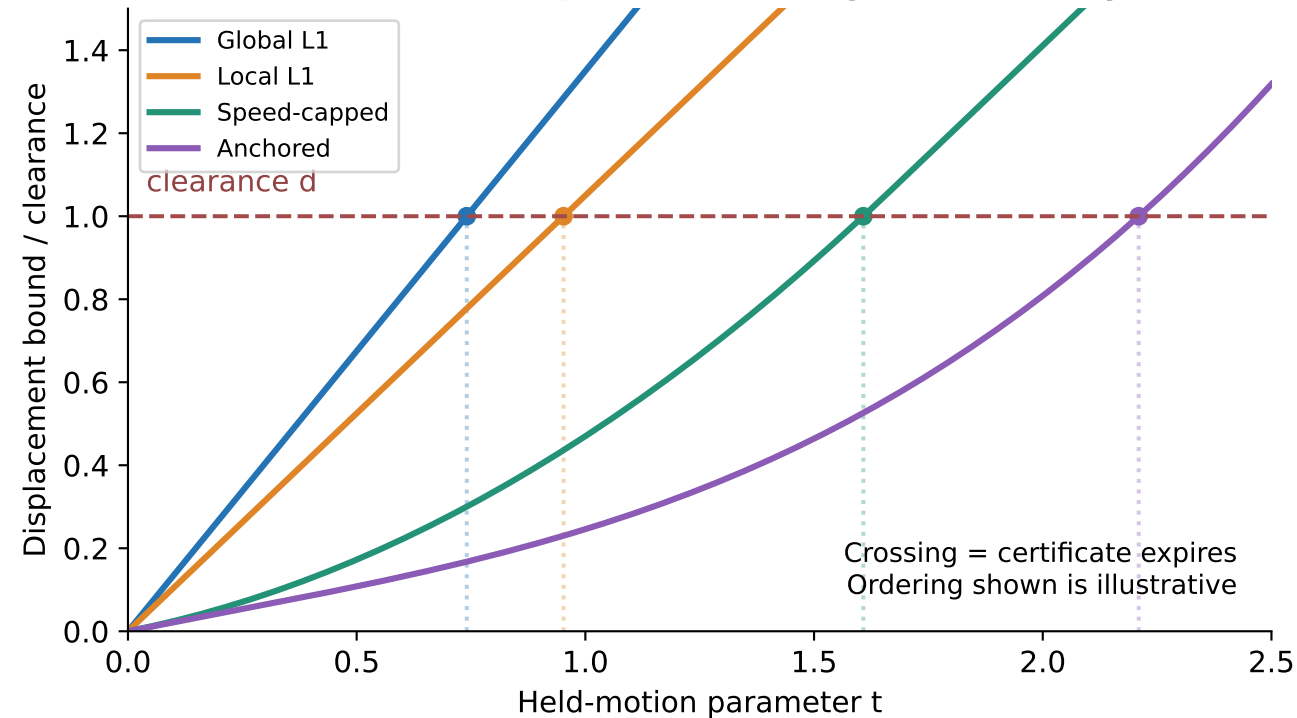
How long can we safely hold a joint-space direction?

A Clearance is a displacement budget

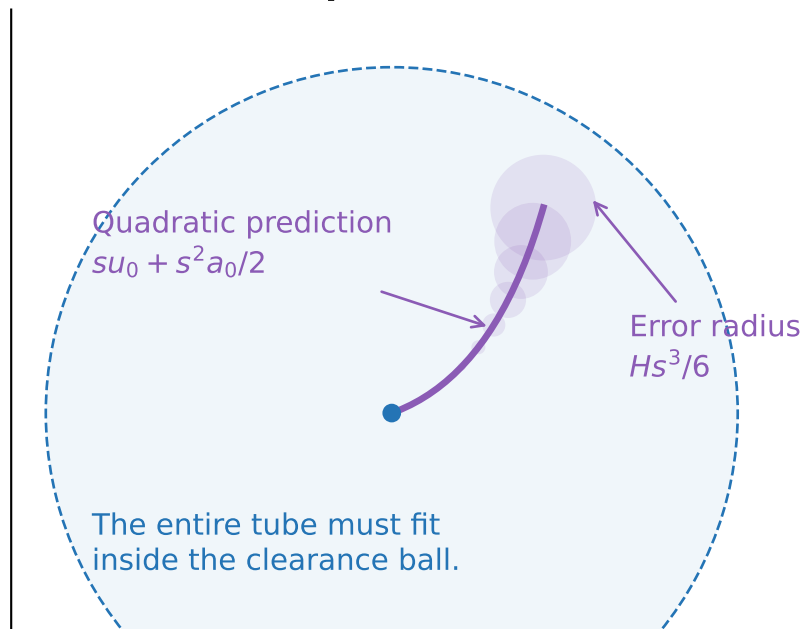


Center motion stays safe while its bound is $\leq d$.
Obstacle inflation includes the robot sphere radius.

B Different bounds spend that budget differently

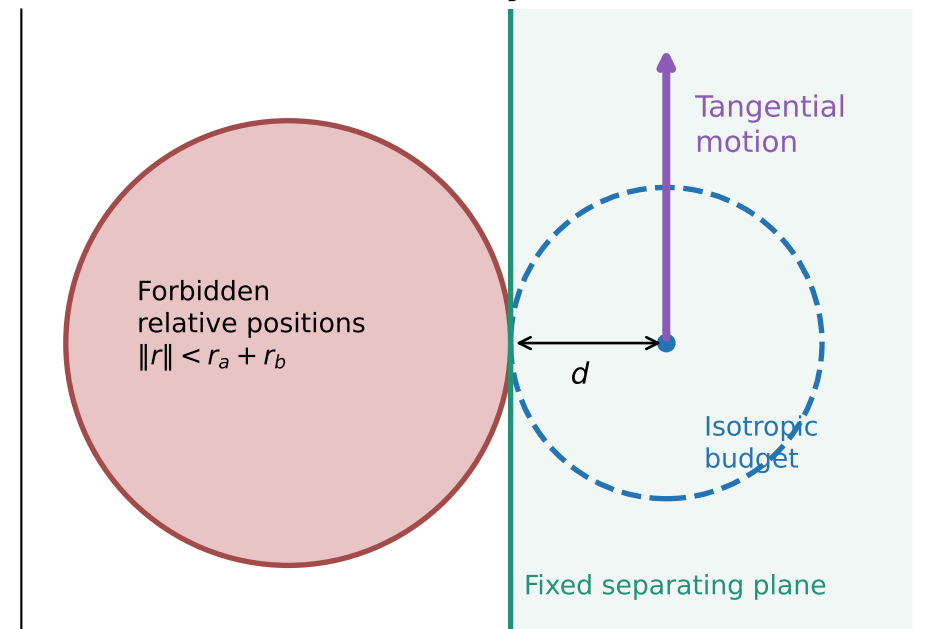


C Anchored: a parabola inside an error tube



Maximize over every $s \in [0, t]$: a safe endpoint is insufficient.

D Self-collision: sideways motion can be free



Normal projection charges only closing motion;
the remainder still accounts for uncertain curvature.

For each collision constraint: take the longest valid certificate. Across constraints: take the shortest, capped by the horizon.
Synthetic illustrations • Certificate expiry is not predicted collision.